Mt. Bigelow 61" (1.55m) telescope / Mont4k CCD Imager Info

- typically, each student group observes about 50% to 75% of a scheduled ASTR302 observing night to acquire images needed to do their project, although a few projects require the whole night. Makeup nights are usually available in case of bad weather on your assigned night.

- the 61" telescope optics and CCD detector are excellent; we also have fast computers, an autoguider, automated dome rotation, and a number of useful observing scripts (eg for scripts to calculate nearly real-time light curve plots) and more. The telescope control room has wifi.

- the Mont4K CCD has excellent QE over much of the optical range CCD, especially in the blue and visual. The QE is much worse in the I filter, where the CCD also exhibits a lot of fringing, making the I filter much less useful for observing:
https://lavinia.as.arizona.edu/~tscopewiki/doku.php?id=public:catalinas:bigelow:kuiper_61:mont4k

- the best targets for the 61"/Mont4k have $11 < V < 16$ and $-5° <$ Declination $< +61°$; brighter targets often saturate the CCD and fainter ones will have lower signal-to-noise (S/N).

- available filters: U, B, V, R, (I) + WFPC2 filters F555W, F606W, F814W,
  and a broadband white light (3300-6100Å) filter, Schott-8612
  To see the filter transmission curves, click on the underlined filter name at
  http://james.as.arizona.edu/~psmith/61inch/FILTERS/filters.html

- the field of view of the Mont4k CCD on the sky is 9.72' x 9.72' ; the CCD has one major bad column plus a vertical seam down the middle between the two halves of the CCD, but is otherwise very clean

- the most commonly used on-chip binning is 3x3 pixels, which produces pixel sizes of 0.43"/pixel on the sky; if better spatial resolution or larger dynamic range is needed, the ccd may be binned 2x2, with gives 0.29"/pixel on the sky

- projects that require relative brightness measurements (ie relative to reference stars within the same image) such as light curves, or targets with known secondary standards within the field-of-view, can be observed over a much wider range of sky conditions than projects which require completely clear skies . The latter are not recommended, since your chances of getting useful data on your observing night are rather small. However, if your project does need absolute brightness measurements,
  1) you will need to select Landolt standard stars in advance and plan when to observe them in between your target observations
  2) have a backup project that you could observe if the sky is not completely clear

- a very useful table to help plan your observations is the MMT Observer's Almanac at
  https://www.mmto.org/wp-content/uploads/2025/11/almanac_2026.pdf
  Scroll down to February and March. All dates and times in the almanac are MST, not UT!

  Each line refers to a single night, telling you when sunset or sunrise will occur, and the times of civil, nautical and astronomical twilight. At the start of the night, the latter three are, respectively, an approximate guide for when to start moving to the first target, when to start focusing the telescope, and when the sky will be as dark as it will get, except for the light from the moon. Times of moonrise and moonset and the moon's phase and brightness on each night are also listed, as well as additional columns that allow you to determine the sidereal time at the start, middle and end of the night.

- Most astronomy websites use Universal dates and times (UT), rather than calendar, or civil, times (MST). UT is the mean time for Greenwich, England, which is 7 time zones earlier than Arizona, so UT 0:00 (ie Greenwich midnight) always occurs at 5:00pm our time. It's important to remember that the UT date for any observing night in Arizona, from 5pm onwards, is always the day after the current calendar date at the start of the night in Arizona!

The astropixels coordinates of the moon for 2026 are listed for UT 00:00 each day, or 5pm MST:
http://astropixels.com/ephemeris/moon/moon2026.html
to repeat: at 5pm each day, the UT date will change to the next calendar day; the civil date in Arizona won't change to the next day until 7 hrs later, at midnight.